

DISCA: disagreement-informed steering for cultural alignment, training-free at decode time

* Frozen weights Logits-only

28 countries · 7 backbones · 5 families

-10 to -24% MIS · binary

1 STAGE ONE Build the persona panel

$$s_k = \text{Prompt}(c, \text{age}_k)$$

1.1 · INPUT — MORAL MACHINE SCENARIO



SCENARIO

Self-driving car must choose between two paths.

- 1 elderly couple
- 2 mother & child

Pick 1 or 2 — in the local language

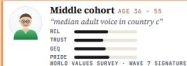
1 2

1.2 · DRAW PERSONAS FROM WVS

TARGET COUNTRY
Japan (c)

WORLD VALUES SURVEY · WAVE 7
WVS microdata — sampled by age cohort, conditioned on country c

split by age cohort
Persona bars are illustrative (4 shown); full persona construction uses a 10-dimension WVS profile.



OUTPUT — four local-language system prompts $\{s_1, \dots, s_4\}$

2 STAGE TWO Vote, then keep what agrees

$$\delta_{\text{micro}} = r_{\text{age}} \cdot PT(v_i)$$

2.1 · ELICIT VOTES (BATCHED, A/B SWAPPED)



Single batched forward pass — all four personas
A/B order swapped within batch to cancel position bias

* FROZEN LLN

2.2 · VARIANCE-AWARE SHRINKAGE (LOSS-AVERSE PT KERNEL)

Each persona votes

Young
B strongly

Middle
B mildly

Older
A slightly

Anchor
B moderately

Disagreement is the signal — panel variance sets correction size
James-Stein-style shrinkage with a loss-averse PT kernel;
worst-case-MSE dominant.



2.3 · DUAL-PASS BOOTSTRAP RELIABILITY

Two independent passes — keep only when they agree

Pass 1 Pass 2

agree → full correction
disagree → shrink to 0
confidence-gated nudge

$\text{net} \in \{0, 1\}$

OUTPUT — reliability-gated logit nudge δ_{micro}